ESAS TERMS

1. What is the metal used to recover copper from a solution of copper sulphate?
   A. Na
   B. Fe*
   C. Hg
   D. Ag

2. Which of the following reason/s why reactions generally proceed faster at higher temperature?
   I. The molecules collide more frequently
   II. The activation energy is less
   A. I only
   B. either I or II
   C. II only
   D. both I and II*

3. What refers to ions that are NOT involved in the overall reactions?
   A. Guest ions
   B. Spectator ions*
   C. Extra ions
   D. Special ions

4. A description of the arrangement of atoms or ions in a material is called.
   A. mixture
   B. composition
   C. component
   D. structure*

5. Materials in this group are composed of one or more metallic elements such as iron, aluminum, copper, titanium, gold, and nickel. What is the group called?
   A. metals*
   B. metalloids
   C. solids
   D. crystals

6. What is the heaviest subatomic particle?
   A. electron
   B. proton
   C. atom
   D. neutron*

7. What is the gas present in the atmosphere which filters out of the sun's ultraviolet light and provides an effective shield against radiation damage to living things?
   A. helium
   B. methane
   C. oxygen
   D. ozone*

8. What percent of human body is carbon?
   A. 25%
   B. 18%*
   C. 15%
   D. 20%

9. In the periodic table of elements, which one of the following is light metal?
   A. Potassium*
   B. Mercury
   C. Silver
   D. Gold

10. Which of the following refers to the diffusion of a solvent into a stronger solution in an attempt to equalize the two concentrations?
    A. Purification
    B. Electrolysis
    C. Osmosis*
    D. Hydrolysis

11. What is the chemical formula of Sulfur Trioxide?
    A. S_3O_2
    B. S_3O
    C. S_2O_3
    D. SO_3*

12. The charge of aluminum atom in the aluminum foil is:
    A. 0*
    B. +1
    C. +2
    D. +3

13. Which device could be used to determine whether a substance is radioactive?
    A. Van de Graaf generator
    B. Geiger counter*
    C. Cyclotron
    D. Linear accelerator

14. What is a substance cannot be decomposed into simpler substance during ordinary chemical reactions?
    A. Atom
    B. Element*
    C. Molecule
    D. Compound

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15. How many moles there in one atom?
    A. 1.66 x 10^-23 moles
    B. 1.66 x 10^-24 moles*
    C. 1.66 x 10^23 moles
    D. 1.66 x 10^24 moles

16. Formula for Aluminum Nitride
    A. Al_2N_2
    B. AlN*
    C. AlN_2
    D. Al_2N

17. All isotopes of the same element have:
    A. Equal no. of protons and neutrons
    B. Equal no. of protons*
    C. The same atomic mass
    D. Equal no. of neutrons

18. With a neutral atom?
    A. the atomic number = the atomic mass
    B. the atomic mass = the mass of neutrons + the mass of protons
    C. the number of protons = the atomic number*
    D. the number of electrons = the number of neutrons

19. What is the common unit used to indicate the mass of a particle?
    A. Atomic mass unit*
    B. Charge unit
    C. Coulomb
    D. Gram

20. Which of the following is an alkaline earth metal?
    A. Ba*
    B. Ti
    C. Li
    D. Na

21. What do you called the horizontal arrangement of elements of increasing atomic number in the periodic table?
    A. period*
    B. ions
    C. valence
    D. none of these*

22. Which of the following metallic element has a meaning "heavy stone"?
    A. Titanium
    B. Iron
    C. Cobalt
    D. Tungsten*

23. Which of the electrostatic forces that exist between ions?
    A. Ionic bond*
    B. Covalent bond
    C. Chemical bond
    D. Atomic bond

24. The sodium ion in sodium chloride, NaCl, has what charge?
    A. +2
    B. -1
    C. -2
    D. +1*

25. Formula for Calcium Chloride
    A. $CaCl_2$*
    B. Ca_2Cl
    C. CaCl
    D. $CaCl_3$

26. Which of these elements has the highest atomic numbers?
    A. Titanium
    B. Uranium
    C. Radium
    D. Plutonium*

27. The following statements are made with regard to the boiling point of a liquid. Which statement is FALSE?
    A. A nonvolatile substance having zero vapor pressure in solution (e.g., sugars or salts) has no true boiling point
    B. The boiling point is the temperature at which the vapor pressure of a liquid equals the applied pressure on the liquid
    C. Combinations of liquids having different boiling points can be separated by slowly raising the temperature to draw off each fraction (i.e., by fractional distillation)
    D. At high elevations, water boils at a lower temperature because of a reduction in the surface tension of the water*

28. How is "molality" defined?
    A. the number of moles of solute in 1000 g of solvent*
    B. the number of moles of solute in 1 L of solution
    C. the number of gram-formula weights of solute per liter
    D. the number of gram-equivalent weights of solute in 1 L of solution

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29. Which of the following elements and compounds is reactive in its pure form?
A. Sodium (Na)*
B. Helium (He)
C. Carbon dioxide (CO_2)
D. Hydrochloric acid (HCl)

30. A steel cannot qualify for stainless prefix until it has at least how many percent of Chromium?
A. 10%*
B. 20%
C. 25%
D. 5%

31. What prefix in steel identification means composition varies from normal limits?
A. E
B. H
C. X*
D. B

32. In preparing a metallographic iron specimen, the grain boundaries are made most visible by which of the following steps?
A. grinding the sample with silicon carbide abrasive
B. polishing the sample with Al_2O_3
C. mounting the sample in an epoxy resin mold
D. etching the sample in a 2% solution of nitric acid in alcohol*

33. Under very slow deformation and at high temperature, it is possible to have some plastic flow in a crystal at a shear stress lower than the critical shear stress. What is the phenomenon called?
A. slip
B. twinning
C. creep*
D. bending

34. In cold working of metals
A. close dimensional tolerance cannot be maintained
B. poor surface finish is obtained
C. recrystallization temperature for steel is reduced
D. strength and hardness of steel is increased*

35. The property of a material which allows it to be drawn into a smaller section is called
A. Plasticity
B. Ductility*
C. Elasticity
D. Malleability

36. What are considered as the building blocks for engineering materials?
A. Atoms*
B. Elements
C. Compounds
D. Matter

37. What is a combination of two or more materials that has properties that the components material do not have by themselves?
A. Compound
B. Composite*
C. Mixture
D. Matrix

38. What physical property of a material that refers to the point at which a material liquefies on heating or solidifies on cooling?
A. Melting point*
B. Curie point
C. Refractive index
D. Specific heat

39. What physical property of a material that refers to the temperature at which ferromagnetic materials can no longer be magnetized by outside forces?
A. Melting point
B. Curie point*
C. Refractive index
D. Specific heat

40. What is a polymer production process that involves forming a polymer chain containing two different monomers?
A. Copolymerization*
B. Blending
C. Alloying
D. Cross-linking

41. Which of the following is a natural magnet?
A. Steel
B. Magnesia
C. Lodestone*
D. Soft iron

42. What is considered as the general purpose, oldest type and widely used cast iron?
A. Gray iron*
B. Malleable iron
C. White iron
D. Ductile iron

43. What is a coal that has been previously burned in an oxygen poor environment.
A. Tuyene
B. Coke*
C. Diamond
D. Hematite

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44. The term brass is used to designate an alloy containing
A. Copper and zinc*
B. Aluminum and iron
C. Copper and aluminum
D. Copper and tin

45. What is the most abundant metal in nature?
A. Aluminum*
B. Steel
C. Iron
D. Copper

46. Galvanized iron is a term referring to iron coated with:
A. aluminum
B. magnesium
C. tin
D. zinc*

47. Which of the following treatments will result in an increase in the fatigue strength of the steel?
A. shot peening*
B. surface roughening
C. cold working
D. annealing

48. What is accomplished by heating raw rubber with small amount of sulfur?
A. polymerization
B. raw rubberation
C. vulcanization*
D. sulfurization

49. Which of these statements is true for twinning?
A. It occurs at lower shear stresses than slip
B. It is the most significant form of plastic deformation
C. It cannot be caused by impact or thermal treatment
D. It frequency occurs in hexagonal close-packed structures*

50. What are Van der Waals forces?
A. weak secondary bonds between atoms*
B. primary bonds between atoms
C. forces between electrons and protons
D. forces not present in liquids

51. Iron is said to be abundant in nature. About how many percent of the earth's crust is iron?
A. 10%
B. 5%*
C. 20%
D. 8%

52. Steels that are used for axles, gears, and similar parts requiring medium to high and strength are known as?
A. Medium-carbon steel*
B. Low-carbon steel
C. Very high-carbon
D. High-carbon steel

53. Polymer comes from Greek words "poly" which means "many" and "meros" which means
A. metal
B. material
C. part*
D. plastic

54. What is the principle behind fiber optics?
A. Reflection*
B. Refraction
C. Diffraction
D. Dispersion

55. The ratio of the velocity of light in vacuum to its velocity of another material is
A. Specific Index
B. Poisson's Ratio
C. Refractive Index*
D. Modulus of Elasticity

56. How tall does a wall mirror have to be in order for you to see all of yourself in the mirror? Imagine you stand a distance "d" from the mirror and your height is "h" and L is the length of mirror required.
A. L = 1/5 h
B. L = 1/4 h
C. L = 1/2 h*
D. L = 1/3 h

57. What is the basic lens formulas?
A. 1/Di = 1/f + Do; So/Si = Do/Di
B. 1/Di + 1/Do = 1/f; So/Si = Di/Do
C. 1/Di = 1/f + Do; Si/So = Do/Di
D. 1/Di + 1/Do = 1/f; So/Si = Do/Di*

58. Which of the following statement/s about sound is/are true?
I. All sounds are produced by the vibration of bodies
II. Sound cannot travel through a vacuum
III. Sounds can be transmitted through some material
A. I and II
B. I, II and III*
C. I and III
D. II and III

59. Objects that emit their own light are called _____ bodies.
   A. solar
   B. luminated*
   C. photovoltaic
   D. illuminated

60. In a mixed gas such as air, the rms speed of different molecules are _____.
   A. proportional to molecular mass
   B. independent of molecular mass
   C. proportional to square root of molecular mass
   D. inversely proportional to the square root of molecular mass*

61. Whenever a body capable of oscillating is acted on by a periodic series of impulses having frequency equal to 1 of the natural frequency of oscillating of the body, the body is set into vibration with a relatively large amplitude. This phenomenon refers to
   A. Doppler's effect
   B. interference
   C. vibration
   D. resonance*

62. When a source of sound or a listener or both in motion relative to the air, the pitch of the sound heard by the listener is not the same as the when source and listener are at rest. This phenomenon refers to?
   A. carrier wave
   B. reverberation effect
   C. delay effect
   D. doppler effect*

63. What is the meaning of an image if the image distance is negative?
   A. The image is real
   B. The image is virtual*
   C. The image is erect
   D. The image is inverted

64. What are the three colors that when mixed equally together will produce white and the combination of any two will produce different colors and intensities?
   A. Red, Blue Green*
   B. Black, Red, Violet
   C. Red, Blue, Yellow
   D. Yellow, Magenta, Cyan

65. Which law states that the ratio of the sines of the angles of incidence and refraction is equal to the ratio of the speeds of light in the two media?
   A. Snell's law*
   B. Doppler's effect
   C. Fourier law
   D. Huygens principle

66. What is an electron volt?
   A. It is a charged electron (with voltage on it) the escaped from its parent atom
   B. It is the English unit of electrical charge of one electron
   C. It is a voltage unit commonly used when measuring the voltage possessed by an electron
   D. It is a unit of energy equal to the energy possessed by an electron accelerating through a potential of one volt*

67. What is LASER stand for?
   A. Light Amplification by Specified Emission of Radiation
   B. Light Amplification by Simultaneous Emission of Radiation
   C. Light Amplification by Stimulated Emission of Radiation*
   D. Light Amplification by Saturated Emission of Radiation

68. The mathematical expression (1 + i)^n is referred to
   A. compounded
   B. interest factor
   C. future factor
   D. compound amount factor*

69. What is the factor name for the formula (1 + i)^-n?
   A. Uniform gradient future worth
   B. Capital recovery
   C. Single payment present worth*
   D. Single payment compounded amount

70. Is the worth of property as shown in the accounting records.
   A. Scrap value
   B. Book value*
   C. Fair value
   D. Use value

71. Is the worth of property which is equal to the original cost less the amount which has been charged to depreciation.
   A. Scrap value
   B. Book value*
   C. Fair value
   D. Use value

72. It is interest earning with equal deposit made at equal interval time.
   A. amortization
   B. dividend
   C. sinking fund*
   D. depletion

73. Is a measure of the effectiveness of an investment of a capital.
A. Rate of return*
B. Effective interest rate
C. Annual worth method
D. Principal or Present worth method

74. The total income equals the total operating cost.
A. Break-even-no-gain-no-loss*
B. In-piece-value
C. Interest rate
D. Scrap value

75. Which of the following corresponds to the meaning of annuity?
A. A lump sum at the end of the year
B. An annual repayment of the load
C. A series of uniform amounts over a period of time*
D. The future worth of a present amount

76. What is an annuity?
A. the future worth of a present amount
B. an annual repayment of a loan
C. a series uniform amounts over a period of time*
D. a lump sum at the end of the year

77. It is the interest earned by the principal which is added to the principal will also earn an interest for the succeeding period.
A. compound interest*
B. simple interest
C. discount
D. interest rate

78. What must two investments with the same present worth and unequal lives have?
A. identical salvage values
B. different salvage values
C. identical equivalent uniform annual cash flows
D. different equivalent uniform annual cash flows*

79. What is the formula for a straight-line depreciation rate?
A. (100% - % net salvage value)/(estimated service life)*
B. (% net salvage value)/(estimated service life)
C. (100% net salvage value)/(estimated service life)
D. (average net salvage value)/(estimated service life)

80. Which of the following is true regarding the minimum attractive rate of return used in judging proposed investments?
A. It is the same for every organization
B. It is larger than the interest rate used to discount expected cash flow from investments
C. It is frequently a policy decision made by an organization's management*
D. It is not relevant in engineering economy studies

81. What is "work in process" classified as?
A. an asset*
B. a liability
C. an expense
D. a revenue

82. Which of the following does NOT affect owner's equity?
A. dividends paid
B. license to start operation*
C. invested capital
D. expense to get license to start operation

83. What is the acid test ratio?
A. the ratio of owner's equity to total current liabilities
B. the ratio of all assets to total liabilities
C. the ratio of current assets (exclusive of inventory) to total current liabilities*
D. the ratio of gross margin to operating, sales, and administrative expenses

84. What is the basic accounting equation ?
A. Assets = liability + owner's equity*
B. Liability = assets + owners' equity
C. Owner's equity = assets + liability
D. Owner's equity = liability – assets

85. The mathematical expression (1 + i) raised to n in Engineering Economics is referred to as the _____.
A. compound value
B. interest factor
C. compound amount factor*
D. future value

86. What do you call the ability to convert assets to cash quickly?
A. Leverage
B. Insolvency
C. Solvency
D. Liquidity*

87. The amount of property in which a willing buyer will pay to a willing seller for the property when neither one is under the compulsion to buy nor sell is called _____ value.
A. Goodwill
B. Second-hand
C. Market*
D. Fair

88. What is the interest accrued over the interest period is also subject to the interest rate in the next period?
A. quarterly interest
B. compound interest*
C. simple interest
D. annual interest

89. Given a free-falling body with air resistance neglected, if the final position of the bod travelled is four times its initial position, by what factor does the velocity of the body increase?
A. twice*
B. half
C. four times
D. same

90. Two sky divers jump at the same time and want to play catch. Where should one aim if he wants the other to catch it?
A. below
B. above because ball falls faster
C. above because ball falls slower
D. straight because there's no air resistance*

91. The path of projectile is:
A. a parabola*
B. an ellipse
C. a apart of a circle
D. a hyperbola

92. A ball is thrown straight up into the air. Ignore air resistance. While the ball is in the air its acceleration.
A. increases
B. is zero
C. remains constant*
D. decreases

93. As a projectile is thrown upward moves in its parabolic path, at what point along its path are the velocity and acceleration vectors for the projectile perpendicular to each other?
A. the launch point
B. the lowest point
C. nowhere
D. the highest point*

94. In dynamics, uniform motion means
A. moving in a straight line
B. having uniform velocity*
C. having constant acceleration
D. moving back and forth

95. As the projectile rises and then falls back to the ground, its acceleration _____.
A. decreases, then increases
B. increases, then decreases
C. increases only
D. remains the same*

96. If the acceleration of a particle is directly proportional to the force acting on it and inversely proportional to the mass of the particle, it follows the Newton's _____ law of motion.
A. first
B. second*
C. third
D. none of these

97. The law pertaining to the collision of two objects in an isolated system, the total momentum of the two objects before collision is equal to the total momentum of the two objects after collision.
A. conservation of energy
B. conservation of momentum*
C. impulse and momentum
D. Newton's second law

98. When an object exerts a force on another object, the second object exerts on the first a force of the same magnitude but in opposite direction. This statement is in accordance to
A. Newton's First Law of Motion
B. Newton's Second Law of Motion
C. Newton's Third Law of Motion*
D. All of these

99. When a force is called a "normal" force, it is
A. the usual force expected given the arrangement of a system
B. a force that is perpendicular to the surface of the Earth at any given location
C. a force that is always vertical
D. a contact force perpendicular to the contact surfaces between two solid objects*

100. Which statement about area of moment of inertia is FALSE?
A. $I = \int d^2\, dA$
B. The parallel axis theorem is used to calculate moments of inertia about a parallel displaced axis
C. The moment of inertia of a large area is equal to the summation of the inertia of the smaller areas within the large area
D. The areas closest to the axis of interest contribute most to the moment of inertia*

101. Newton's first law of motion says:
A. every action has an equal but opposite reaction
B. energy cannot be created or destroyed
C. gravitational force is inversely proportional to the square of the separation distances
D. the momentum of a particle remains unchanged in the absence of unbalanced force*

102. The field of study that describes the effect of forces acting on bodies at rest or in motion?
A. Strength of Materials
B. Mechanics of Materials
C. Theory of Structures
D. Engineering Mechanics*

103. What is the SI unit of force?
A. Joule
B. Watt
C. Newton*
D. Dyne

104. Rolling resistance is due to
A. friction between body and surface
B. change in linear momentum
C. change in angular momentum
D. deformation of the rolling body and the surface*

105. A cable carrying a constant uniform load per unit length with respect to the horizontal axis will have the shape of a _____.
A. straight line
B. parabola*
C. catenary
D. hyperbola

106. What is the principle works in the hydraulic brakes of an automobiles?
A. Bernoulli's Law
B. Pascal's Law*
C. Archimedes' Principle
D. Posieuille's Law

107. What is the classification of a turbine if the water moves along the vanes toward the axis of rotation of the runner?
A. Axial flow
B. Radial flow*
C. Parallel flow
D. Mixed flow

108. "An external pressure exerted on a fluid is transmitted uniformly throughout the volume of the fluid". This statement is known as
A. Bernoulli's Theorem
B. Pascal's Principle*
C. Archimedes' Principle
D. Torricelli's Theorem

109. What can you say about the following statement regarding centrifugal pumps?
"The discharge of the centrifugal pumps can be throttled without the building up an excessive pressure or overloading its driving unit."
I. True II. False
A. either I or II
B. both I and II
C. II
D. I*

110. Which law that for a confined fluid, the pressure at a point has the same magnitude in all directions?
A. Pascal's Law*
B. Bernoulli's Theorem
C. Avogadro's Law
D. Amagat's Law

111. Which of the following flow meters measure(s) the average fluid velocity rather than a point or local velocity in a pipe?
I. Venturi meter
II. Pitot tube
III. Impact tube
IV. Orifice meter
V. Hot-wire anemometer
A. I only
B. II only
C. I and IV*
D. II and V

112. Which of the following is TRUE for a propeller-type impeller centrifugal pump?
A. The head developed is due to the push of the vanes and flow is almost entirely axial*
B. The head developed cannot be determined by the rotation of the vanes
C. The head developed is due to the rotation of the impeller
D. The head developed is due to the action of centrifugal force

113. What is the device used to determine the specific gravity of the liquids?
A. Psychrometer
B. Hydrometer*
C. Manometer
D. Venturi Meter

114. Which of the following statement is FALSE?
   A. Steady flows do not change with time at any point
   B. The Navier-Stokes equation is the equation of motion for a viscous Newtonian fluid
   C. Bernoulli's equation only holds on the same streamline
   D. The Reynolds number is the ratio of the viscous force to the internal fore*

115. Fluid like oil, gasoline, water, alcohol are Newtonian fluids.
   A. Charles' Law
   B. True*
   C. False
   D. Boyle's Law

116. Ideal fluid has:
   A. no surface tension
   B. no surface tension, no viscosity
   C. no surface tension, no viscosity and no compressibility*
   D. surface tension, viscosity and compressibility

117. Commercially used pressure measuring device today is
   A. Bourdon Tube Gauge*
   B. Manometer
   C. Kelvin Gauge
   D. Barometer

118. Cavitation is the result of
   A. static pressure in a fluid becoming less than fluid vapor pressure*
   B. exposure of concrete to salt water
   C. heat treatment of a low carbon steel
   D. improper welding technique

119. What is force per unit area?
   A. Work
   B. Energy
   C. Pressure*
   D. Power

120. This principle states that "Where the velocity of a fluid is high, the pressure is low, and where the velocity is low, the pressure is high".
   A. Archimedes Principle
   B. Torricelli's Theorem
   C. Flow Principle
   D. Bernoulli's Principle*

121. Which of the following is the bases for Bernoulli's law of fluid flow?
   A. principle of conservation of mass
   B. principle of conservation of energy*
   C. continuity equation
   D. Colebrook equation

122. Hydraulic gradient of a pipeline represents is
   A. pressure head*
   B. velocity head
   C. datum head
   D. total head

123. Which statement is true for a fluid?
   A. It cannot sustain a shear force
   B. It cannot sustain a shear force at rest*
   C. It is a liquid only
   D. It has a very regular molecular structure

124. Which of the following must be satisfied by the flow of any fluid, real or ideal?
   I. Newton's second law of motion
   II. the continuity equation
   III. the requirement of a uniform velocity distribution
   IV. Newton's law of viscosity
   V. the principle of conservation of energy
   A. I, II and III
   B. I, II, and IV
   C. I, II, and V*
   D. I, II, III and IV

125. Surface tension has which of the following properties?
   I. It has units of force per unit length
   II. It exists whenever there is a density discontinuity
   III. It is strongly affected by pressure
   A. I only
   B. II only
   C. III only
   D. I and II*

126. Which of the following describes a streamline?
I. It is a mathematical concept
II. It cannot be crossed by the flow
III. It is a line of constant entropy
A. I only
B. II only
C. I and II*
D. I and III

127. Which of the following ratio represents a physical interpretation of the Reynolds number?
A. Re = buoyant forces/inertial forces*
B. Re = viscous forces/inertial forces
C. Re = drag forces/viscous forces
D. Re = inertial forces/viscous forces

128. The hydraulic radius is
A. the mean radius of the pipe
B. the radius of the pipe bend on the line
C. the wetted perimeter of a conduit divided by the area of flow
D. the cross-sectional fluid area divided by the wetted perimeter*

129. A non-viscous incompressible liquid is flowing through a horizontal pipe of constant cross section. Bernoulli's equation and the equation of continuity predict that the drop in pressure along the pipe:
A. is zero*
B. depends on the length of the pipe
C. depends on the fluid velocity
D. depends on the cross-sectional area of the pipe

130. A non-viscous incompressible fluid is pumped steadily into the narrow end of a long tapered pipe and emerges from the wide end. The pressure at the input is greater than at the output. A possible explanation is:
A. the fluid speed increases from input to output
B. the fluid speed is the same at the two ends
C. the fluid is flowing uphill*
D. the fluid is flowing downhill

131. Which of the following refers to the parallel axis theorem for second moment of area for strength of materials?
A. Maxwell theorem
B. Young's theorem
C. Steiner's theorem*
D. Mohr's theorem

132. What is another terms for modulus of rigidity?
A. Modulus of elasticity
B. Young's modulus
C. Bulk modulus
D. Shear modulus*

133. What refers to the point in which bending moment changes sign through a zero value in strength of materials?
A. Zero stress point
B. Critical point
C. Point of contraflexure*
D. Point of inflection

134. Which of the following material has the least modulus of elasticity?
A. Copper
B. Aluminum
C. Steel
D. Glass*

135. The modulus of rigidity of a steel member is _____.
A. defined as the length divided by the moment of inertial
B. defined as the unit shear stress divided by the unit shear deformation*
C. equal to the modulus of elasticity divided by one plus Poisson's ratio
D. a function of the length and depth

136. Under which of the following type of loading does the fatigue occurs?
A. repeated load*
B. plane load
C. static load
D. high load

137. Modulus of rigidity of a steel is
A. Define as the unit shear stress divided by unit shear deformation (shear strain)*
B. Equal to the modulus of elasticity divided by one plus Poisson's ratio
C. Defined as the length by the moment of inertia
D. Equal to approximately 7/10 of the modulus of elasticity

138. What is the unit of strain?
A. Pa
B. unitless*
C. N-m
D. N-m/s

139. What is stress?
A. force per unit length
B. mass per unit acceleration
C. force per unit area*
D. force per unit volume

140. The simple mathematical statement of the relationship between elastic stress and strain: stress is proportional to strain. What is this law?
A. Boyle's law
B. Hooke's law*
C. Charles' law
D. Gas law

141. What is the definition of normal strain, ϵ? (δ is elongation, and L is the length of the specimen).
A. ϵ = (L + δ)/L
B. ϵ = (L + δ)/δ
C. ϵ = δ/(L + δ)
D. ϵ = δ/L*

142. What is the maximum moment of a beam supported at both ends and carries a uniform load of w throughout its entire length?
A. wL/8
B. wL2/4
C. wL/2
D. wL2/8*

143. The maximum bending moment of a beam simply supported at the both ends and subjected to a total load W uniformly distributed over its length L is expressed by the formula.
A. WL/8*
B. WL^2/8
C. WL/2
D. WL/4

144. The maximum bending moment in a simple supported beam of span L and carrying a concentrated load "W" at beam's midpoint.
A. WL
B. WL/2
C. WL/4*
D. 2WL

145. A beam of rectangular cross-section is to be cut from a circular log of diameter D. What should be the ratio of the depth of the beam to its width for maximum strength in pure bending?
A. sqrt. of 2*
B. (sqrt. of 2)/2
C. sqrt. of 3
D. (sqrt. of 3)/3

146. Which of the following are true statements about the modulus of elasticity, E?
A. It is the same as the rupture modulus
B. It is the slope of the stress-strain diagram in the linearly elastic region*
C. It is the ratio of stress to volumetric strain
D. Its value depends only on the temperature of the material

147. what is the proper relationship between the modulus of elasticity, E, the Poisson's ratio v, and the bulk modulus of elasticity, K?
A. E = K(1 – 2v)
B. E = K(1 – v)
C. E = 3K/(1 – 2v)
D. E = 3K(1 – 2v)*

148. Where do stress concentrations occur?
I. near the points of applications of concentrated loads
II. along the entire length of high distributed loads
III. at discontinuities
A. I and II
B. I and III*
C. II and III
D. I, II and III

149. In a stress-strain diagram, what is the correct term for the stress level at ϵ = 0.2% offset?
A. the elastic limit
B. the plastic limit
C. the offset rupture stress
D. the offset yield stress*

150. A specimen is subjected to a load. When the load is removed, the strain disappears. From this information, which of the following can be deducted about this material?
A. It is elastic*
B. It is plastic
C. It has a high modulus of elasticity
D. It does not obey Hooke's law

151. Which of the following may be the Poisson ratio of a material?
A. 0.35*
B. 0.52
C. 0.55
D. 0.60

152. Structural steel elements subjected to torsion develop
A. tensile stress
B. shearing stress*
C. compressive stress
D. moment

153. For a bending beam under transverse (bending) loading, where is the location of maximum shear stress?
   A. at the top edge
   B. at the bottom edge
   C. at the neutral axis*
   D. at a location between the top edge and the neutral axis

154. The deflection of beam is
   A. directly proportional to the modulus of elasticity and moment of inertia
   B. inversely proportional to the modulus of elasticity and length of the beam cube
   C. inversely proportional to the modulus of elasticity and moment of inertia*
   D. inversely proportional to the weight and length

155. The slenderness ratio of a column is generally defined as the ratio of
   A. unsupported length to its length radius of gyration*
   B. length to its minimum width
   C. unsupported length to its maximum radius of gyration
   D. unsupported length to its maximum cross-section area

156. "Modulus of resilience" is:
   A. the same as the modulus of elasticity
   B. a measure of a materials ability to store strain energy*
   C. the reciprocal of the modulus of elasticity
   D. a measure of the strain induced in a member due to a change in temperature

157. The stress in an elastic material is:
   A. inversely proportional to the material's yield strength
   B. inversely proportional to the force acting
   C. proportional to the displacement of the material upon by the force*
   D. proportional to the length of the material subject to the force

158. Which of the following formula is INCORRECT for temperature conversion?
   A. Tf = 5/9 Tc + 32
   B. Tf = 9/5 Tc – 32
   C. Tf = 9/5 Tc + 32*
   D. Tf = 5/9 Tc – 32

159. An adiabatic process is characterized by which of the following?
   A. It is reversible
   B. The heat transfer is zero*
   C. it is isothermal
   D. The entropy change is zero

160. An open system of First Law of Thermodynamics should be utilized for all the following EXCEPT _____.
   A. a turbine
   B. a piston-cylinder device with no inlet/exhaust valves*
   C. a compressor
   D. a nozzle

161. Which of the following in the thermodynamics is considered as the heat content of a system?
   A. Internal heat
   B. Molar heat
   C. Entropy
   D. Enthalpy*

162. What law of nature underpins the concept of a balanced chemical equation?
   A. Gay-Lussac law
   B. Faraday's law
   C. Second law of Thermodynamics*
   D. First law of Thermodynamics

163. What law states that for all adiabatic processes between two specified states of a closed system, the network done is the same regardless of the nature of the closed system and the details of the process?
   A. Fourth Law of Thermodynamics
   B. First Law of Thermodynamics
   C. First Law of Thermodynamics*
   D. Third Law of Thermodynamics

164. Which of the following does NOT transfer work to or from a system?
   A. An electrical resistance heater
   B. A moving piston
   C. The expanding membrane of the balloon
   D. A membrane that burst*

165. In a combustion chamber, fuel is burned to raise the temperature of the medium prior to production of work through an expansion process. The heat generated during the combustion reaction is the _____.
   A. heat of formation
   B. specific heat
   C. heat of vaporization
   D. heat of reaction*

166. Which of the following would be identified with a control volume in thermodynamics?
A. Expansion of the gas in a cylinder after combustion
B. Compression of air in a cylinder
C. Filling a tire with air at a service station*
D. The Goodyear blimp during flight

167. What is the transfer of heat due to motions of a fluid near the surface of an object?
A. insulation
B. radiation
C. convection*
D. conduction

168. What refers to the measure of the disorder present in a given substance or system?
A. Enthalpy
B. Entropy*
C. Heat capacity
D. Molar heat

169. A rigid container is heated by the sun. There is NO shaft work associated with the container. From the first law of thermodynamics, what is the resulting work equal to?
A. the change in internal energy
B. zero*
C. the heat transfer
D. the volume times the change in pressure

170. Relationship between temperature and pressure at constant volume.
A. Henry's law
B. Charles' law
C. Gay-Lussac's law*
D. Boyle's law

171. Which of the following are intensive properties?
I. temperature
II. pressure
III. composition
IV. mass
A. I only
B. IV only
C. I and II
D. I, II and III*

172. Which of the following thermodynamic relations is INCORRECT?
A. T dS = dU + p dV
B. T dS = dH – V dp
C. U = Q – W*
D. H = U + pV

173. What is the value of the work done for a closed, reversible, isometric system?
A. zero*
B. positive
C. negative
D. positive or negative

174. During an adiabatic, internally reversible process, what is true about the change in entropy?
A. It is always zero*
B. It is always less than zero
C. It is always greater than zero
D. It is infinite

175. The ideal reversible Carnot cycle involves several basic processes. What type of processes are they?
A. all adiabatic
B. all isentropic
C. two adiabatic and two isentropic
D. two isothermal and two isentropic*

176. The temperature at which vapor pressure of a liquids equal to the atmospheric pressure is?
A. distillation
B. surface tension
C. boiling point*
D. vaporization

177. What is the temperature when water and vapor are in equilibrium with the atmospheric pressure?
A. Ice point
B. Steam point*
C. Critical point
D. Freezing point

178. Entropy is the measure of
A. the change in enthalpy of a system
B. the internal energy of a gas
C. the heat capacity of a substance
D. randomness or disorder*

179. Enthalpy of an ideal gas is a function of:
A. internal energy
B. entropy
C. temperature*
D. pressure

180. A device used to keep moisture from passing through the system is called
A. trap
B. aerator
C. humidifier
D. dehydrator*

181. The ideal, reversible Carnot cycle involves four basic processes. What type of processes are they?
A. all isothermal
B. all adiabatic
C. two adiabatic and two isentropic
D. two isothermal and two isentropic*

182. An isentropic process:
A. Is adiabatic and reversible*
B. Is reversible by may not be adiabatic
C. Is adiabatic but may not be reversible
D. Is always reversible

183. It state that if two bodies are in thermal equilibrium with a third body, they are also in thermal equilibrium with each other.
A. first law of thermodynamics
B. zeroth law of thermodynamics*
C. third law of thermodynamics
D. second law of thermodynamics

184. Which of the following statement is FALSE?
A. Interrupts are used to develop RTOS
B. RTOS performs tasks in predictable amount of time
C. Kernel is the one component of any OS
D. Windows 98 is RTOS*

185. The number of address lines requires to address a memory of size 32 K is _____ lines.
A. 18
B. 16
C. 13
D. 15*

186. What is the standardized communication technique for expressing instructions to a computer?
A. flow chart
B. programming language*
C. algorithm
D. logic

187. HTTP:// is an identifier that appears at the beginning of most web page. What does HTTP stands for?
A. Hypertime Text Process
B. Hypertext Transfer Protocol*
C. Hypertext Transmission Process
D. Hypertext Time Protocol

188. Which of the following portion of hard disk space served as RAM?
A. Cache
B. Random Memory
C. Read-Only Memory
D. Virtual Memory*

189. Which of the following are used in designing and documenting complex processes or programs that help to visualize what is going on and thereby help the viewer to understand a process or algorithm?
A. Charts
B. Diagrams
C. Flowcharts*
D. Networks

190. Which of the following is a correct IF statement?
A. IF (I,GE,5)
B. IF (I.GE.5)
C. IF (I,GE,5) GO TO 7
D. IF (I.GE.5) GO TO 7*

191. Encryption can provide all of the following features of information systems security EXCEPT _____.
A. confidentiality
B. integrity
C. authenticity
D. privacy*

192. What is the UPS stand for?
A. Uninterrupting Power Supply
B. Uninterruptible Power Supply*
C. United Power Supply
D. Uninterruptible Portable Supply

193. The Central Processing Unit of computer is a combination of _____.
A. arithmetic logic and input unit
B. control and storage
C. arithmetic logic and control unit*
D. control and input unit

194. A technician has installed a new video card and the computer will now only boot into VGA mode. Which of the following may be the cause of the problem?
A. There is an operating system incompatibility
B. The external monitor is not compatible
C. The new card is not DVI compatible
D. The incorrect drivers are installed*

195. Which of the following computer program is NOT considered open-source software?
A. Firefox
B. Microsoft Office*
C. Linux
D. OpenOffice.org

196. When installing the heatsink back into a computer after replacing a failed processor, which of the following must be done during installation?
   A. Thermal compound must be installed between the processor and the heatsink*
   B. Thermal compound must be installed between the processer and the system board
   C. An isopropyl alcohol pad must be placed between the processor and the heatsink
   D. An isopropyl alcohol pad must be placed between the processor and the system board

197. Which of the following do you called the brain of any computer system?
   A. Arithmetic Logic Unit
   B. Central Processing Unit*
   C. Random Access Memory
   D. Control Processing Unit

198. A printed circuit board that has (at least) slots to connect cards into. Often, they also include CPU and memory is called ____.
   A. mainframe
   B. IDE port
   C. motherboard*
   D. server

199. Which of the following is the basic symbol used as a beginning or termination on logic diagram or flowcharts?
   A. square box
   B. oval symbol*
   C. rectangular box
   D. diamond box

200. What form of software can be also called firmware?
   A. cache memory
   B. operating system
   C. system BIOS*
   D. random memory

201. In printing your document in the computer, what is the default mode?
   A. page view
   B. landscape
   C. page set-up
   D. portrait*

202. Which of the following is an instruction that causes data to be brought from memory into an accumulation register?
   A. LOAD*
   B. LOOP
   C. FETCH
   D. JUMP

203. IEEE 1284 is a standard communication that defines:
   A. Wireless communication
   B. Wired communication
   C. Parallel communication*
   D. Serial communication

204. A network of private computers infected with malicious software and controlled as a group without owner's knowledge like sending spam messages.
   A. Botnet*
   B. Phishing
   C. Hacking
   D. Trojan horse

205. A software program designed to enter a computer system, usually a network, through security "holes" and then replicate itself.
   A. Computer Virus
   B. Worm*
   C. Trojan Horse
   D. HTTP

206. An abbreviation for binary digit is
   A. byte
   B. bid
   C. bit*
   D. bin

207. A byte is composed of how man bits?
   A. 2
   B. 4
   C. 6
   D. 8*

208. 1 kB is how many bytes?
   A. 1,000
   B. 1,024*
   C. 1,012
   D. 1,048

209. In digital codes, how do you identify codes running from 1 to 10 or 0 to 9?
   A. Decimal
   B. Standard base 10*
   C. Gray
   D. Octal

210. When a binary point of a binary number is moved one place to the right, what happen to its value?
   A. divides by 2
   B. multiplies by 2*
   C. increased by 10
   D. multiplies by 10

211. Boolean algebra is essentially based on
A. symbols
B. logic*
C. truth
D. numbers

212. ASCII means?
A. American Standard Code for Information Input
B. American Standard Code for Input Information
C. American Standard Capacity for Information Input
D. American Standard Code for Information Interchange*

213. What is the standard method for coding data on 80-column, 12-row card?
A. ASCII
B. EBCDIC code
C. Hollerith code*
D. Baudot code

214. Octal coding involves grouping the bits in
A. 6's
B. 9's
C. 1's
D. 3's*

215. What is called the group of bits, that may be manipulated or stored at one time by the CPU.
A. program
B. address
C. word*
D. software

216. What is the programmable, multi-use machine that accepts data-raw facts and figures and processes, or manipulates it into information we can use, such as summaries or totals.
A. Cash Registration
B. Floppy disk
C. Adding Machine
D. Computer*

217. _____ are round pieces of flat plastic that store data and program as magnetized spots.
A. Hard disk
B. Floppy disk*
C. Compact disk
D. CD-ROM

218. This is a secondary storage medium, generally non-removable disk made out of metal and covered with a magnetic recording surface. It holds data in the form of magnetized spots and are usually built into the computer system cabinet.
A. Hard disk*
B. Floppy disk
C. Compact disk
D. CD-ROM

219. A _____ is a list of instructions that the computer must follow in order to process data into information.
A. Program*
B. Pseudocode
C. Flowchart
D. File

220. It is a process of using microcomputer, mouse scanner, laser printer that combines highly digital text and images to produce a quality output.
A. CAM
B. Desktop Publishing*
C. CAD
D. CAE

221. A software that allows users to have fast and flexible access to large information and documents.
A. internet
B. e-mail
C. hypertext*
D. hyperlink

222. A network of networks where the people can retrieve information, communicate, entertain, do remote purchasing from the comfort of a personal computer is called.
A. Internet*
B. E-mail
C. Fax
D. Home TV shopping

223. It refers to a network in which all microcomputers and other communication devices are connected to a central hub, such a file server or host computer.
A. Star Network*
B. Ring Network
C. Bus Network
D. Hybrid Network

224. It refers to a network in which all microcomputers and other communication devices are connected in a continuous loop.
A. Star network
B. Ring network*
C. Bus network
D. Hybrid network

225. A parity bit is also called _____.
A. check bit*
B. RAM
C. silicon type
D. driver

226. A two-way parallel path which connects the processor and memory containing addresses.
A. cache memory
B. control bus
C. address bus*
D. buffer

227. Which of the following is the fastest method to transfer data wirelessly?
A. infrared
B. Bluetooth
C. 802.11*
D. GPRS

228. Which of the following is NOT a computer programming language?
A. FORTRAN
B. COBOL
C. LOTUS*
D. PASCAL

229. What WWW stands for?
A. World Wide Weber
B. World Wide Web*
C. World Wide Width
D. World Wide Working

230. MS-Word is a type of what software?
A. System software
B. Operating system
C. Application software*
D. System architecture

231. MS-DOS is a _____ software.
A. system*
B. operating
C. application
D. system architecture

232. What is the symbol used in decision making?
A. oval
B. rectangle
C. diamond*
D. square

233. What does FTTH mean?
A. Flexible to the House
B. Flexible to the Home
C. Fiber to the Home*
D. Fiber to the House

234. An instruction that causes data in the accumulator to be moved to the memory or a peripheral register.
A. FETCH
B. STORE*
C. LOOP
D. LOAD

235. Which one is the portion of an instruction cycle where the instruction is sent from memory to the instruction register?
A. LOAD
B. ACCULULATE
C. EXECUTE
D. FETCH*

236. What is the system program used to translate directly an assembly language to machine language called?
A. Assembler*
B. Complier
C. Interrupter
D. Debugger

237. _____ is a software that allows users to have fast and flexible access to information in large documents, constructing associations among data items needed.
A. Voice mail
B. Internet
C. E-mail
D. Hypertext*

238. What is VIRUS means,
A. Vital Information Resource Under Siege*
B. Very Important Resource Under Siege
C. Vulnerable Information Reserved Under Siege
D. Vital Information Reserved Under Siege

239. ALU means,
A. Asian Labor Union
B. Arithmetic Logic Unit*
C. American Loggers Union
D. Arithmetic Logic Universe

240. BIOS means,
A. Base Input Output System
B. Basic Input Output System*
C. Basic Input Output Society
D. Basic Input Output Surplus

241. USB means,
A. Universal Serial Bus*
B. Universal Series Bus
C. Universal Sentry Bus
D. Universal Sentinel Bond

242. What refers to a technique where each user of a computer takes turn, under the control of the operating system, using the entire computer main memory for a certain length of time?
A. multitasking
B. time sharing*
C. handshaking
D. batch processing

243. A computer virus that infects the macros that are attached to documents and spreadsheet is called,
A. micro virus*
B. file virus
C. document virus
D. spreadsheet virus

244. Which of the following is not pre-defined variable in MATLab?
A. pi
B. inf
C. j
D. gravity*

245. In MATLab, this keyword immediately moves to the next iteration of the loop:
A. update
B. goto
C. continue*
D. break

246. This MATLab commands clears all data and variables stored in memory:
A. clc
B. clear*
C. delete
D. deallocate

247. Which of the following is used to see if two elements are equal in MATLab?
A. !-
B. ==*
C. isequal
D. =

248. The clc command is used to
A. clear the command window*
B. erase everything in the mfile
C. clean the desktop
D. save the existing mfile

249. Which refers to the establishing interrelationships between people and things in such a way that human and material resources are effectively focused toward achieving the goal of the company?
A. planning
B. leading
C. organizing*
D. controlling

250. If you are appointed as a division manager, which of the following task is MOST likely you will do?
A. determine the resources needed
B. develop strategies and tactics
C. set a standard
D. set goals*

251. Which of the following technique must a manager use if he decides to absorb the risk in the project?
A. Use a different technology
B. Buying insurance
C. Use a different supplier
D. Create buffer in the form of management reserve or extra time to schedule*

252. Engineers can become good managers only through _____.
A. Experience
B. Taking master degree in management
C. Effective career planning*
D. Trainings

253. Which of the following is usually the last step in the process of performing a project?
A. Implementing a plan
B. Evaluate project success
C. Monitor and control the project*
D. Select appropriate performance measures

254. Buying an insurance is a form of risk _____.
A. elimination
B. sharing*
C. reduction
D. absorption

255. Which of the following is not used in the Program Evaluation and Review Technique?
A. Construction of projects
B. Computer programming assignments
C. Preparation of bids and proposal
D. Queueing problems*

256. When an engineer enters management, what is the most likely problem that he/she finds difficult to acquire?
   A. Learning to trust others
   B. Learning how to work through others
   C. Learning how to take satisfaction in the work of others
   D. All of the above*

257. What is the most essential attribute of a project manager?
   A. Leadership*
   B. Charisma
   C. Communication skills
   D. Knowledge

258. A committee companies or corporations created for a short term purpose only is called _____ committee.
   A. interim
   B. buffer
   C. temporary
   D. ad hoc*

259. It is defined as a course of action aimed at ensuring that the organization will achieve its objectives.
   A. Goal
   B. Program
   C. Strategy*
   D. Plan

260. The technique used by a manager used when evaluating alternative using qualitative evaluation is called _____ technique.
   A. comparison
   B. intuition and subjective judgement*
   C. rational
   D. analytical

261. The process of determining how specific task can best be accomplished on time with available resources is called _____.
   A. operational*
   B. strategic
   C. skilled
   D. organizational

262. A form of departmentalization in which everyone engages in one functional activity, such as engineering or marketing, is grouped into one unit is called _____ organization.
   A. functional*
   B. product
   C. matrix
   D. technical

263. Which of the following is an advantage of free-rein style of leadership?
   A. Time consuming and cost ineffective
   B. Little ideas from subordinate in decision-making
   C. Leader managerial control and high degree of risk*
   D. All of the above

264. The function of an engineer where the engineer undertakes the activity of tuning an product concept to a finished physical term is called _____.
   A. research
   B. design and development*
   C. manufacturing
   D. testing

265. It is measured by the amount of resources that a manager can allocate without the need to get an approval from his or her manager.
   A. Responsibility
   B. Authority*
   C. Leadership
   D. Trade-off skill

266. One of the mnemonic management tool used is the SMEAC. What does the acronym SMEAC stands for?
   A. Situation, Mission, Execution, Administration, Coordination
   B. Situation, Mission, Execution, Application, Communication
   C. Situation, Mission, Execution, Administration, Communication*
   D. Strategy, Mission, Execution, Administration, Communication

267. In project management, what is usually the first step underlying in the process of performing a project?
   A. Select appropriate performance measures
   B. Define the goals of the project and their relative importance
   C. Identify a need for a product or service*
   D. Develop a technological concept

268. In the process of performing a project, after the need for a product or service is identified, what is usually the next step?
   A. Define the goals of the project and their relative importance*
   B. Develop a budget
   C. Develop a schedule
   D. Develop the technological concept

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269. What is usually the last step in the process of performing a project?
A. Select appropriate performance measures
B. Implementing a plan
C. Monitor and control the project*
D. Evaluate project success

270. To consider the consequence of uncertainly on project management, laws on project management are developed. One of which is "A careless planned project will take _____ times longer to complete than expected".
A. three*
B. four
C. two
D. two and a half

271. In the typical functional organization hierarchy, the chief engineer is under the _____.
A. Finance manager
B. General manager*
C. Manufacturing manager
D. Marketing manager

272. What is a diagram of the organization's official positions and formal lines of authority called?
A. Organization chart*
B. Policy chart
C. Authority chart
D. Control chart

273. What refers to the activity of incorporating the technical know how with the ability to organize and coordinate workforce, materials, equipment and all other resources including money?
A. Engineering management*
B. Engineering technology
C. Technical manager
D. General management

274. What network model enables engineer managers to schedule, monitor and control large and complex projects by using only one time factor per activity?
A. Forecasting
B. Critical path method*
C. Program evaluation review technique
D. Simulation

275. What network model enables engineer managers to schedule, monitor and control large and complex projects by employing three time estimate for each activity?
A. Forecasting
B. Critical path method
C. Program evaluation review technique*
D. Simulation

276. For a project manager to have an effective means of identifying and communicating the planned activities and their interrelationships, he must use a network technique. One of the network techniques is commonly known as PERT. What does PERT stands for?
A. Project evaluation review technique
B. Program evaluation review technique*
C. Path evaluation review technique
D. Program execution review technique

277. What is the standard color for safety signages to indicate the presence of radiation hazards example in the X-ray laboratory in the hospital?
A. green with combination of red
B. orange
C. blue
D. purple with combination of yellow*

278. This provides detailed information about a specific hazardous material and contains some of the following such name of substance physical hazards, health hazard, safe handling procedures, control measures, emergency first aid procedures, etc. What is this called?
A. Management Safety Duty Sheet
B. Material Signages Data Sheet
C. Management Safety Data Sheet
D. Material Safety Data Sheet*

279. Which of the following is/are hazardous practice/s?
A. using equipment improperly
B. failure to warn or secure
C. horseplay
D. all of these*

280. What is the reason why incidents are investigated?
A. to prevent or control a future recurrence on a similar incident*
B. to determine the causes of incident
C. to gather evidence of the incident
D. to determine who is at fault in the incident

281. Which of the following is NOT a hazardous practice?
A. using the right tools/equipment*
B. servicing equipment in motion
C. using improvised tools
D. operating without authority

282. When analyzing the evidence you want to do the following EXCEPT,
A. review equipment manuals
B. review employee medical records*
C. review witness statements
D. review company standard operating procedure

283. Your investigation tool bag should include the following EXCEPT
A. digital camera
B. tape recorder*
C. graph paper
D. tape measure

284. An incident is considered to be an
A. accident
B. near miss
C. accident and near miss*
D. none of these

285. Most accidents in the workplace can be forecasted
A. True*
B. False
C. Maybe True or False
D. Neither True nor False

286. Which of the following information cannot be found in MSDS?
A. first and measures
B. accident release measures
C. handling and storage information
D. medicine to be taken by the effected person if exposed to the chemical*

287. The Chairman and member of the Board of Electrical Engineers are appointed by _____.
A. Secretary of Department of Labor and Employment
B. Chairman of the Professional Regulation Commission
C. President of the Philippines*
D. Secretary of Department of Energy

288. Based in Section 18 of the R.A. 7920, the minimum age requirement for qualification of applicants for Registered Master Electrician Examinations is at least _____ years of age.
A. eighteen*
B. nineteen
C. twenty-one
D. twenty

289. Which of the following is qualified as the Chairman of Continuing Professional Development Council for Electrical Engineering?
A. Representative from Association of Professional Organization
B. PRBEE board member*
C. Representative from academy
D. PRC Commissioner

290. Section 23 of Article III of RA 7920 – Issuance of Certification of Registration. The professional is required to renew every ____ year/s upon payment of the registration fee.
A. three*
B. two
C. six
D. five

291. What is the term used when an American citizen is allowed to practice his profession, e.g., electrical engineering profession, here in the Philippines?
A. APEC registry
B. ACPE registry
C. Foreign Reciprocity*
D. ASEAN Engineer

292. Which of the following is a ground for disqualification of the applicant for REE licensure examination?
A. has been finally convicted by the court of an offense involving moral turpitude*
B. good reputation with low moral values
C. citizen of the Philippines
D. at least twenty-one (21) years of age

293. In R.A. 7920, which of the following is not considered as a practice of electrical engineering?
A. manufacturing of electrical equipment
B. teaching in secondary schools*
C. preparing electrical plans
D. sales and distribution of electrical equipment

294. The relative percentage weight of the subject ESAS in the REE board exam is _____ percent.
A. 25
B. 30*
C. 35
D. 40

295. The passing general weighted average rating for the licensure examination for Registered Electrical Engineers must be 70% with NO GRADE below _____ in any subjects in Mathematics, ESAS, and Electrical Engineering as required under Section 19(b) of RA 7920.
A. 55%
B. 65%
C. 60%
D. 50%*

296. Vacancies of the Board seat shall be filled by the _____ from list of recommmendees selected by the Commissioner who was chosen from the list of nominees submitted by the integrated and accredited association for the unexpired term only.
A. IIEE
B. President*
C. Chairman of the Board
D. Commissioner

297. The terms of office of every Board member including the Chairman is _____ years.
A. 6
B. 4
C. 3*
D. 5

298. Why is it that the electrical system of aircraft is not under the jurisdiction of Electrical Engineers?
A. Electrical Engineers are prone to accident
B. Frequency is too high for them to handle
C. Machines in the aircraft are intended for Aeronautical Engineers
D. Frequency is under the scope of Avionics*

299. What does PRBEE stands for?
A. Philippine Regulating Board of Electrical Engineering
B. Professional Regulatory Board of Electrical Engineering
C. Philippine Regulatory Board of Electrical Engineering*
D. Professional Regulating Board of Electrical Engineering

300. Which are NOT included in duties and responsibilities of PRBEE?
A. Supervise and regulate the practice of electrical engineering in the Philippines
B. Issue special permits to individual foreign electrical engineers and electricians for specific projects*
C. Perform such other functions and duties as may be necessary to implement effectivity of this Act
D. All of these

301. Which one is NOT requirement for the applicant to apply for registration as a registered electrical engineer?
A. at least twenty-one (21) years of age
B. citizen of the Philippines
C. has not been finally convicted by the court of an offense involving moral turpitude
D. good reputation with low moral values*

302. This refers to energy-consuming equipment including motors, heaters, furnaces, light sources and other device which utilize electrical energy, for any purpose.
A. Industrial plant
B. Utilization equipment*
C. Processing plant
D. Transformer

303. The Electrical Engineering shall observe such a system of moral principles and values as a set of dynamic standards to guide his conduct and way of life. It is his _____ to practice his professional according to this Code of Ethics.
A. manner
B. right
C. way
D. duty*

304. The Electrical Engineer shall observe such system of _____ principles and values as a set of dynamic standards to guide his conduct and way of life. It is his duty to practice his profession according to this Code of Ethics.
A. dignity
B. moral*
C. basic
D. engineering

305. Requirements for examination as an RME is Rule _____ of RA 7920.
A. 15
B. 16
C. 17*
D. 14

306. RA 7920 Rule 2(f) defining?
A. Commission
B. RA 184, old Electrical Engineering Law*
C. Commissioner
D. CPE

ENGR. CLIBOURN

307. _____ refers to the installed capacity of an alternating current electric plant or supply equipment, or the connected load of industrial plants/commercial building per Section 2(h) Article I RA 7920.
A. Computed load
B. Total connected load
C. Electric plant
D. Kilo-Volt Ampere*

308. In R.A. 7920, Commission refers to:
A. PRC*
B. PRBEE
C. IIEE
D. DOLE

309. Branch circuit conductors supplying a single motor shall have an ampacity of how many percent of its rated full-load current?
A. 100%
B. 115%
C. 150%
D. 125%*

310. The clearance between conductors and roof where the service conductors are passing over roofs must be _____ mm.
A. 1,000
B. 1,500
C. 2,000
D. 2,500*

311. In house wiring, ground conductors must not be _____.
A. insulated
B. identified
C. placed in a circuit
D. fused*

312. Which of the following is CORRECT regarding the sizing of the rating of the overcurrent protective device of a circuit?
A. 125% of the non-continuous load plus 80% of the continuous load
B. 125% of the continuous load plus 80% of the non-continuous load
C. 125% of the continuous load plus 100% of the non-continuous load*
D. 125% of the non-continuous load plus 100% of the continuous load

313. The usual function of a disconnect switched in high voltage circuits is to
A. open the circuit in the event of an overload
B. open or close the circuit under load
C. maintain continuity of service
D. isolate from energized busses, equipment which are not in service*

314. "The earth" best defines of which of the following term?
A. bonding
B. ground electrode
C. ground*
D. grounded conductor

315. What is the MAXIMUM distance from the flooring to the receptacle outlet in dwelling unit in mm?
A. 2,500
B. 1,800*
C. 1,500
D. 2,800

316. Where receptacles are connected to a 30-A branch circuit, the maximum allowable cord and plug connected load shall NOT be more than _____ A.
A. 24*
B. 26
C. 30
D. 34

317. In Rigid Metal Conduit (RMC) wiring, conduits shall be supported at least every _____ mm.
A. 1,500
B. 2,000
C. 3,000*
D. 1,000

318. The total load current on overcurrent device located in a panelboard shall NOT exceed a certain percentage of its rating. What is this percentage?
A. 120%
B. 90%
C. 100%
D. 80%*

319. How many times the diameter of the largest raceways shall the length of the box be in straight pulls?
A. 6
B. 4
C. 8*
D. 10

320. The Philippine Electrical Code Part 1 2009 was based on the National Electrical Code 2005 while the Philippine Electrical Code Part 1 2017 Edition was based on the National Electrical Code _____.
A. 2017*
B. 2011
C. 2008
D. 2014

ENGR. CLIBOURN

321. Which of the following is/are hazardous practice/s?
A. using equipment improperly
B. failure to warn or secure
C. horseplay
D. all of these*

322. What is the reason why incidents are investigated?
A. To prevent or control a future recurrence on a similar incident*
B. To determine the causes of incident
C. To gather evidence of the incident
D. To determine who is at fault in the incident

323. Which of the following is NOT a hazardous practice?
A. using the right tools/equipment*
B. using improvised tools
C. service equipment in motion
D. operating without authority

324. Which of the following insulation color identify the grounded conductor of a branch circuit?
A. black
B. green
C. yellow
D. white*

325. What is the MAXIMUM number of overcurrent devices for a lighting and appliance that can be installed in a single cabinet?
A. 36
B. 42
C. 24
D. 48*

326. The Philippine Electrical Code is passed on the Board of Electrical Engineers last November, 2017. The Philippine Electrical Code 2017 is Board Resolution number ____ series of 2017.
A. 14
B. 16
C. 20
D. 18*

327. Which locations may be temporarily subject to dampness or wetness?
A. Dry*
B. Damp
C. Moist
D. Wet

328. Rule used to determine sizes and rating of circuit conductors and devices.
A. manufacturing rule
B. construction rule
C. wiring design rule*
D. installation rule

329. Motors must be labeled with the following: nameplate rating, manufacturer's name and etc. This is:
A. manufacturing rule*
B. construction rule
C. wiring design rule
D. installation rule

330. The conductor used to connect the grounding electrode to the equipment grounding conductor, to the grounded conductor, or to both, of the circuit of the service equipment or at the service of a separately derived system
A. Grounded Conductor
B. Grounding Conductor
C. Equipment Grounding Conductor
D. Grounding Electrode Conductor*

331. For continuous load, the maximum current is expected to continue for ____.
A. 2 hours
B. 4 hours
C. 6 hours or more
D. 3 hours or more*

332. What is the minimum branch circuit rating with continuous and non-continuous load?
A. 125%, 100%*
B. 125%, 80%
C. 120%, 80%
D. 120%, 100%

333. The minimum size of conductors to be used for lighting purposes is
A. 2.0 mm^2*
B. 5.5 mm^2
C. 3.5 mm^2
D. 8.0 mm^2

334. Branch circuit conductors supplying a single motor shall have an ampacity of how many percent of its rated full-load current?
A. 100%
B. 115%
C. 150%
D. 125%*

335. The PEC Part 1 2017 Edition is intended as a design specification or an instruction manual to qualified person.
I. True II. False
A. Both I or II
B. Either I or II
C. II only
D. I only*

336. The clearance between conductors and roof where the service conductors are passing over roofs must be _____ mm.
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D. 3 hours or more*

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A. 125%, 100%*
B. 125%, 80%
C. 120%, 80%
D. 120%, 100%

359. The minimum size of conductors to be used for lighting purposes is
A. 2.0 mm^2*
B. 5.5 mm^2
C. 3.5 mm^2
D. 8.0 mm^2

360. For hermetic motor compressor, how many percent of the full load current is the allowable rating of the branch disconnecting means?
A. 125
B. 115*
C. 150
D. 100

361. The minimum size of copper equipment conductor required for equipment connected to a 40 Ampere circuit is?
A. 14.0 mm^2
B. 8.0 mm^2
C. 5.5 mm^2*
D. 22.0 mm^2

362. Which of these is not a factor in the selection of fuse rating for protection of a branch circuit?
A. Branch full-load current
B. System Impedance*
C. Branch fault-current
D. System voltage

363. Which of the following is not a standard over-current protection device rating?
A. 50
B. 1500*
C. 225
D. 150

364. In estimating the loading of a branch circuit, what loading shall be used for each receptacle?
A. 160 VA
B. 120 VA
C. 180 VA*
D. 150 VA

365. Receptacle rating of a non-motor load for a 40 ampere branch is.
A. 20
B. 30
C. 40*
D. 50

366. What do you call a device which is also known as Earth Leakage Circuit Breaker (ELCB)?
A. RCB
B. RCD
C. GFCI*
D. MCB

367. What is derating factor in each of 7 to 9 conductors in a raceway?
A. 80%
B. 85%
C. 70%*
D. 75%

368. What is the current carrying capacity of each conductor if 36-8.0 sq. mm THWN copper conductors are install in an auxiliary gutter?
A. 20 A each*
B. 40 A each
C. 50 A each
D. 45 A each

369. What is the maximum number of conductor in a raceway that does not need to be derated?
A. 2
B. 3*
C. 4
D. 5

370. Branch circuit larger than _____ ampere shall supply only non-lighting outlet load?
A. 20
B. 50*
C. 15
D. 30

371. Which of the following is not a standard classification for a branch circuit supplying several loads?
A. 30 A
B. 20 A
C. 40 A
D. 25 A*

372. Find the maximum vertical clearance of the receptacle outlet from the floor?
A. 2,500 mm
B. 1,800 mm*
C. 3,700 mm
D. 4,000 mm

373. Floor receptacle outlets are not counted unless they are within a distance of __ mm from the wall.
A. 350
B. 400
C. 450*
D. 500

374. For 120 to 277 volt, nominal to ground shall be permitted to supply which of the following?
A. None of these
B. Screw shell devices with loads
C. Lighting fixture
D. Auxiliary equipment of electric discharge ampere*

375. When the temperature of the wire increases, what happens to the resistance?
A. Varies
B. Remains the same
C. Increases*
D. Decreases

376. Grounding allows ___ opposition path of current flow.
A. Low*
B. High
C. Effective
D. Medium

377. The minimum diameter of a steel ground electrode is
A. 10 mm
B. 12 mm
C. 14 mm
D. 16 mm*

378. A single electrode consisting of a rod, pipe, or plate that does not have a resistance to ground of ___ ohms or less than shall be augmented by one additional electrode.
A. 20
B. 30
C. 25*
D. 35

379. In Philippine Electrical Code (PEC), what is the maximum distance between bus way supports?
A. 1,000 mm
B. 1,500 mm*
C. 1,800 mm
D. 2,000 mm

380. What is the maximum number of quarter bends permitted for a single run of rigid metallic conduit between outlets?
A. 4*
B. 5
C. 2
D. 3

381. Which is not a standard distribution voltage in the Philippines?
A. 13.2 kV
B. 13.8 kV
C. 34.5 kV
D. 69 kV*

382. Where must the surge arrester be located?
A. Away from the transformer
B. Near the circuit breaker
C. Near the air break switch
D. Near the transformer*

383. According to the Philippine Electrical Code at an ambient temperature of 30°C, 8.0 mm^2 copper conductors with the following insulation have the following ampacities: TW insulation – 30 A; THW insulation – 45 A; for THHN insulation – 50 A. If each of these insulated wires carries the same 40 amperes which will generate heat the fastest?
A. TW insulated wire*
B. THW insulated wires
C. Each will generate heat at the same rate
D. THHN insulated wire

384. Vertical clearance from ground. Service-drop conductors where not in excess of 600 volts, nominal shall have the minimum clearance from final grade of __ over residential property and those commercial areas not subject to truck traffic where the voltage does not exceed 300 volts to ground.
A. 3,100 mm
B. 3,700 mm*
C. 4,600 mm
D. 5,500 mm

385. Luminaire type permitted in clothes closets.
A. Incandescent fixture with open ampere
B. Incandescent fixture with partially enclosed ampere
C. Pendant luminaire
D. Surface mounted or recessed incandescent fixture with a completely enclosed lamp*

386. Which of the following luminaire types are NOT permitted in clothes closets?
I. Incandescent fixtures with open lamps
II. Incandescent fixtures with partially enclosed lamps
III. Pendant luminaire
IV. Surface mounted or recessed incandescent fixture with a completely enclosed lamp
A. I, II and III*
B. I, II and IV
C. I, III and IV
D. II, III and IV

387. Medium voltage cable insulations is rated for ___ volts or higher.
A. 600
B. 1,000
C. 230
D. 2,001*

388. In dwelling units with a hallway of 3 m or more in length, at least how many receptacle outlet/s shall be installed?
A. 3
B. 2
C. 1*
D. None

389. A 4-wire system has a high-leg voltage that is indicated by a wire of color___.
A. Black
B. Orange*
C. Brown
D. Red

390. The color reserved for use in equipment grounding color is
A. Yellow
B. Green*
C. White
D. Gray

391. Which of the following insulation color identify the grounded conductor of a branch circuit?
A. Black
B. Green
C. Yellow
D. White*

392. What is the maximum operating temperature of type UF conductor?
A. 75°C
B. 90°C
C. 60°C*
D. 110°C

393. A conducting connections, whether intentional or accidental, between an electric circuit or equipment and the earth, or to some conducting body that serves in place of the earth.
A. Grounded
B. Ground*
C. Short
D. Fault

394. Which of the following is NOT permitted by the PEC under certain conditions for grounding?
A. Isolated
B. Unguarded*
C. Insulated
D. Guarded

395. What is a protective device designed to protect an individual against electric shock?
A. Circuit Breaker
B. AFCI
C. Differential Relay
D. GFCI*

396. A conduit bend needed if the conduit end is running through an outlet box for connection of lighting fixture, convenience outlet, etc.
A. Gooseneck bend
B. Saddle bend
C. Corner bend
D. Offset bend*

397. In PEC 1, where does a bonding jumper is connected?
A. Near distribution panel
B. Load side of service drop
C. Equipment enclosure*
D. Transformer enclosure

398. What is the maximum distance between two outlets in dwelling units in mm?
A. 1,500
B. 2,000
C. 1,800*
D. 1,600

399. What is the metallic tubing used for insulated conductor?
A. Steel conduit*
B. Tray
C. Flexible conduit
D. Trunking

400. Is ground can be intentional or accidental?
I. True II. False
A. I only*
B. II only
C. Both I and II
D. Either I or II

401. All grounding connections are made _____ and electrically _____.
A. temporary, non-continuous
B. permanent, continuous*
C. temporary, continuous
D. permanent, non-continuous

402. The size of the sole connection of a grounding electrode conductor connected to a concrete-encased electrode shall not be required to be which of the following?
A. Connected to an electrode encased by at least 100 mm of concrete
B. Connected to an electrode encased by at least 50 mm of concrete
C. Larger than 22 mm^2 copper*
D. Larger than 38 mm^2 copper

403. A refrigerator receptacle can be fed from a:
I. 15 – ampere individual branch circuit
II. 20 – ampere individual branch circuit
III. 20 – ampere small appliance branch circuit
A. II and III
B. II only
C. III only
D. I, II and III*

404. The lighting load for an office building expressed in terms of a unit load in volt-amperes per square meter must be at least
A. 24
B. 32
C. 28*
D. 16

405. Which of the following is the electrical device that is usually required to be installed in moisture area in the dwelling unit in order to protect the person from electric shock?
A. Circuit Breaker
B. Arc-Fault Circuit Interrupter
C. Power Fuse
D. Ground-Fault Circuit Interrupter*

406. The lighting load for a dwelling expressed in terms of a unit load in volt-amperes per square meter must be at least _____.
A. 24*
B. 12
C. 16
D. 28

END

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